Federated Transfer Learning For Diabetic Retinopathy Detection Using CNN Architectures

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***Abstract*—Diabetic Retinopathy is a complication of diabetes that could cause vision loss or even blindness if not detected in the early stages. As a result, having a regular eye exam is critical for maintaining a healthy retina and preventing damage to the eyes. Due to the lack of ophthalmologists in developing countries, there should be a faster approach to analyze the condition of these fundus images collected by different opticians. In this case, developing deep learning approaches comes in handy to enhance the effectiveness of Diabetic Retinopathy (DR) diagnosis accurately. Our main motivation is to develop a system that could manage various medical institutions. The critical part is to preserve privacy while training such deep learning models, which Federated Learning enables a decentralized training method by only sharing the parameters not the actual data. This study investigates three main models using standard transfer learning, Federated Averaging, and Federated Proximal frameworks. We demonstrate that our three models, including standard, FedAVG, and FedProx, were able to detect DR or non-DR images with the accuracy of 92.19%, 90.07%, and 85.81% respectively.**

***Index Terms*—Federated Learning, Transfer Learning, Medical Imaging, Diabetic Retinopathy**

I. Introduction

According to the statistics from the International Diabetes Federation (IDF), the number of people with diabetes was 557 million in 2021. It is predicted to reach more than 630 million by the end of this decade [1]. In another report by the Centers for Disease Control and Prevention in 2020, more than 10% (34.2 million) of Americans have diabetes, of which 7.3 million are undiagnosed [2]. People with diabetes are at increased risk of eye diseases such as Diabetic Retinopathy, diabetic macular edema (DME), etc. [3]. DR, the most common eye disease related to diabetes, is caused by damage to the small blood vessels in the retina. If left untreated, these bloody retinal veins can cause a variety of vision loss and even blindness [4]. Some typical signs of DR among diabetic people could include microaneurysms, hard exudates, vitreous hemorrhage, etc. [5]. Retinal images for people dealing with DR could be defined in 5 main stages: normal, mild, moderate, severe, and proliferative [6]. Since DR does not show symptoms at the first stages, early diagnosis could have a vast clinical, social, and even economical importance in dealing with this disease. In fact, not only early detection could prevent vision loss, but it could also be cost-effective compared to a situation where the patient is dealing with blindness. However, there could be a lack of ophthalmologists in developing countries. In this case, there may be a high demand for such ophthalmologists to diagnose DR on time properly. Consequently, researchers have deployed several types of deep learning techniques to enable faster diagnosis of this critical disease [7].

One major adopted research area in deep learning uses Convolutional Neural Networks (CNNs), which have demonstrated great results for all image processing tasks, especially image classification [8]. Basically, a CNN network contains different convolution layers responsible for consecutively feeding the input image within layers. Along with the convolution layers, CNNs have different filters and pooling layers, all as a feature learning part. The second part consists of fully connected layers which perform the classification. However, training efficient CNNs for specific tasks could take massive time and computation costs. To solve this, scientists started training various types of CNNs on large famous datasets resulting in creation of pretrained CNN architectures such as AlexNet [9], VGGNet [10], ResNet [11], etc. Employing these pretrained architectures enables scientists to transfer knowledge gained from other images. Also, using this technique prevents scientists from collecting large image datasets, which could be time-consuming or even unavailable. This method, called Transfer Learning (TL), deploys the pretrained model. It fine-tunes the parameters at the classification part based on the target dataset and its characteristics. There have been various approaches using CNN architectures for classifying DR images, which we discuss in the related works. In the proposed approach, we have utilized the AlexNet architecture as the main part of our training. The models transfer all the learnings from AlexNet to be deployed for DR training and classification.

Although traditional centralized machine learning models have demonstrated efficient results, but deploying these models in real life may not be applicable. This is because of the strict privacy regulations which limits the use of data, especially in healthcare domain. The HIPAA (Health Insurance Portability and Accountability Act of 1996) is an instance of regulations which aims to protect the sensitive information of patient by allowing the use of data only with the consent of the patient [12]. In this case, the data is not allowed to be shared with the models for training purposes. This is where Federated Learning (FL) could play a key role by deploying a decentralized training method where the data is not shared. FL attempts to preserve privacy by only sharing the parameters not the actual data. In addition to the scope of our study, applying deep learning models is not possible in real-time, where the clinicians can upload retina images into their end-devices to check if the patients' eyes are at considerable risk. As a result, we simulate this procedure by applying our models on five publicly available DR datasets, where the FL model can be trained without access to data.

The remaining part of this paper is organized as follows: Section II provides a literature review of the conducted research studies on transfer learning and federated learning. The proposed methods and utilized datasets are explained in Section III. Then, we discuss our results and discussions in Section IV. Conclusions are finally given in Section V.

II. Related Work

Transfer Deep Learning. When it comes to healthcare, available data is minimal and hard to access. As a result, training complicated models is not applicable. However, using pre-trained models comes in handy to solve this issue. The Transfer Learning technique provides more efficient models based on specific tasks. Recent studies have demonstrated outstanding potential for training CNN models on medical imaging data, especially DR images.

Masood and Luthra [13] deployed transfer learning for classifying eye images using InceptionV3. The authors trained their model using balanced data gathered from a publicly available dataset, named EyePACS [14]. The images were preprocessed using three tasks of downsampling, subtraction, and cropping. These techniques were performed to highlight the key features of the DR images. Authors demonstrated that subtracting the local average color improved their model by achieving an accuracy of 48.2% [13].

In another work, authors [15] deployed transfer learning using two types of CNNs, including AlexNet and GoogLeNet. The authors adopted to publicly available datasets from Eye-Pacs [16] and Messidor-1 [17]. Prior to training the models, authors preprocessed the data using Otsu's method [18] and performed different augmentation techniques to improve the models' outcomes. The highest accuracy of the proposed transfer learning models for classifying DR images was 74.5%.

Similarly, Li et al. [19] attempted to classify DR images collected from DR1 [20] and Messidor-1 [17] datasets. They deployed three pre-trained models as a transfer learning approach, including AlexNet, VGG-s, VGGNet-16, and VGGNet-19. To fine-tune the pre-trained models, authors performed two approaches, every layer fine-tuning and connected layers' fine-tunning. Among all the models mentioned earlier, VGG-s demonstrated the highest accuracy within both datasets. [19].

Le et al. [21] also proposed a method for DR detection using Optical Coherence Tomography (OCT) images. They employed VGG-16 as their architecture for classifying OCT images based on healthy, no DR, and DR classes. Adopting the transfer learning with a cross-validation method enabled authors only to retrain the last nine layers within their architecture to achieve the highest accuracy of 87.27% [21]. They showed that deploying transfer learning could assist with the lack of experienced ophthalmologists.

To detect DR at early stages, authors in [22] investigated various deep transfer learning models, including AlexNet, Res-Net18, SqueezeNet, GoogleNet, VGG-16, and VGG-19. They utilized the Asia Pacific Tele-Ophthalmology Society (APTOS) 2019 dataset [23] for training and testing the performance of their models. Additionally, the authors applied data augmentation techniques (such as reflection around different axes) to achieve better performance on the testing set. Among all the proposed models, AlexNet demonstrated the highest accuracy of 97.9% with the least training time and computation costs [22].

In a similar approach, Gangwar and Ravi [24] proposed a hybrid transfer learning model using a CNN architecture called Inception-ResNet-v2. They modified the architecture by adding a custom block containing four convolutional layers, filters, and dropouts. Messidor-1 and APTOS datasets were utilized for evaluating the model's performance, which achieved the accuracy of 72.33% and 82.18% respectively.

In a recent CNN-based Diabetic Retinopathy detection study [25], AlexNet, GoogLeNet, and ResNet50 frameworks were retrained using the transfer learning with retina images from several datasets including EyePACS, Messidor, IDRiD, and University of Auckland datasets. They investigated the effect of using images from the single, cross, and multiple datasets and observed that training deep networks as much diverse images from different datasets improves the DR detection accuracy as 94.6%.

Federated Learning. A new research paradigm, called Federated Learning, was first mentioned in a work conducted by McMahan et al. [26]. The main idea behind this technique is to train machine learning models in a privacy-preserving manner. Unlike traditional centralized machine learning models, FL aims to train the models without seeing the data. This is performed by training a global model generated and updated using the local models' parameters. Consequently, FL could provide efficient training on multiple devices or institutions, where accessing to data is not allowed or limited [27], [28]. Healthcare institutions could be an excellent fit for applying FL, where confidential data is not accessible [29]. As a result, we aim to deploy FL for detecting DR images. Herein, we discuss the conducted research studies on FL and DR together.

Data heterogeneity has been one of the main challenges in training FL models in real-time applications. Since independent and identically distribution of data is not practical among healthcare institutions, Qu et al. [30] explored the impacts of this challenge on three popular FL approaches (FedSGD, Federated Averaging (FedAVG) [26], and Cyclical weight transfer (CWT) [31]). They proposed their approach on classifying two medical tasks, including Alzheimer's and DR. The authors employed three main optimizations to overcome the adverse effects of different types of data heterogeneity (quantity, label distribution, and imaging acquisition) on models' performance.

The weighted average, weighted loss, and batch normalization are the optimizations employed by the authors. They showed the improvements of each model by comparing it with a centralized model, where models are faced with [30].

In another approach, authors in [32] tackled data heterogeneity by utilizing transformers instead of CNNs along with their FL setup. They employed two proposed methods of Vision Transformers (ViT) and CNN-Transformer hybrid (ViT-H) [33]. For the FL setup, the authors applied FedAVG [26] and CWT [31]. The performance of different combinations of above techniques were evaluated using two datasets (EyePACS [14] and CIFAR [34]). Along with all the models, ViT-H with the FedAVG setup achieves the highest prediction accuracy of 83% for the retina dataset. Authors showed that employing transformers within FL could enhance the effectiveness of models achieving a more suitable global model [32].

Although FL was designed to preserve privacy, there are still scenarios where data privacy could be threatened. Deploying Differential Privacy (DP) [35] with FL could enhance privacy by adding random noise within the computed gradients. Malekzadeh et al. [36] developed a system called Dopamine to enable training Deep Neural Networks (DNN) using Fl and DP. The authors utilized SqueezeNet as their DNN architecture, trained on the APTOS dataset to classify DR images. In comparison with other approaches, authors showed that Dopamine could enhance the privacy of medical data [36].

Lastly, a DR classification in an FL environment was conducted by Lo et al. [37]. As a part of their work, they employed four clients for their FL setup to enable collaboration among the medical institutions. The authors utilized OCT and OCT angiography (OCTA) images to detect referable and non-referable DR classes. The model used a pre-trained architecture, VGG-16, to enable better feature extraction. Overall, the proposed model achieved the highest accuracy of 88.8%, and the authors demonstrated comparable results for deploying FL for medical applications [37].

III. Methods

We propose three main models for classifying retina images based on two classes of non-DR and DR. Non-DR stands for patients whose eyes are in a good and healthy form. On the other hand, the DR class demonstrates patients who need emergency care and treatments. This section discusses our models, namely Transfer Deep Learning, Federated Averaging (FedAVG), and Federated Proximal (FedProx) algorithms. Then, we will describe the utilized datasets for training and testing the proposed models.

A. Transfer Deep Learning

Our standard model performs as our only non-FL approach for comparison purposes. It is a traditional deep learning approach, where the data transfers to the model without any limitations. This method is one of the most frequent applications in research-based studies. Additionally, we have used transfer learning to create a model for detecting DR cases with comparable performance. Among numerous types of CNNs, we employ AlexNet architecture, which has demonstrated efficient results for various tasks in medical imaging. AlexNet is a pre-trained network containing weights from the ImageNet dataset. Since the current AlexNet is sufficient for our training purposes, we only modify the last three fully connected layers. In this case, the models can predict our two classes effectively.

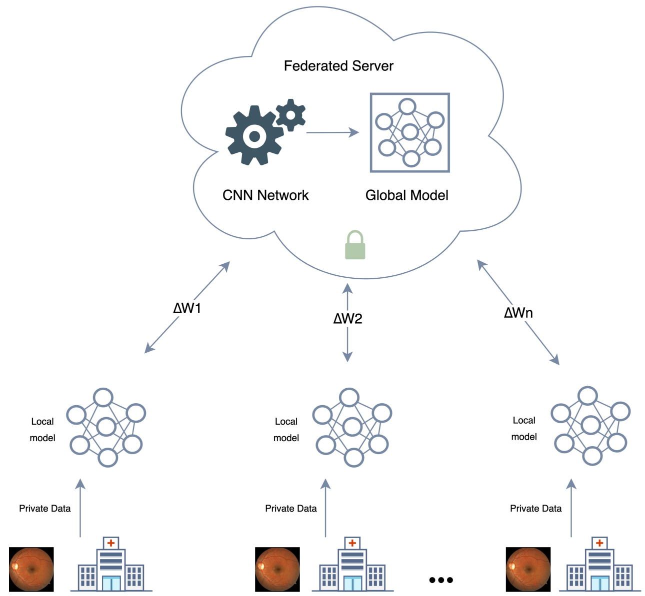

Fig. 1. Federated Learning Concept in Healthcare

B. Federated Learning

Federated Learning aims to train clients in a decentralized manner, where the data stays on the end-user instead of being shared with the model. A Federated Learning setup could be summarized as depicted in Fig. 1. First, a global model is sent to various clients holding confidential data. Next, every client starts training its local model using its data. These local models contain weights and parameters after the training is completed. Then, the server asks for the trained local models to be sent to the server. After all the models are received, an aggregation algorithm is used to combine the parameters from the local models. In this case, a global model is built without seeing the data. This could be repetitive for many rounds until the expected performance is achieved. Two types of FL aggregation algorithms are deployed as two separate models in our proposed approach. Federated Averaging algorithm (FedAVG) [26] is one of the most popular aggregation strategies. FedAVG averages all the parameters within local models received from clients. Federated Proximal Algorithm (FedProx) [38] is a variation of FedAVG, which aims to improve the performance of federated networks facing heterogeneity challenges. It assigns a proximal term, which takes the local updates closer to the global model more seriously. Both of our FL aggregation strategies use the same transfer learning technique as the standard model. This means

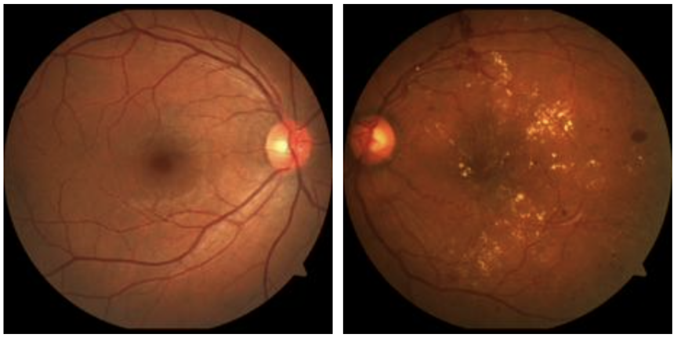

Fig. 2. Normal (left), Critical (right)

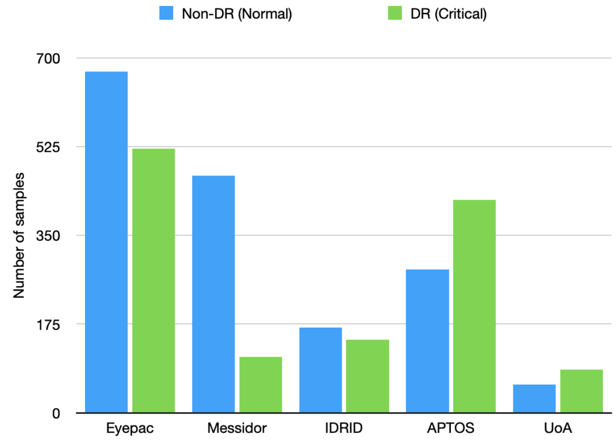

Fig. 3. Number of samples for datasets

that the global model contains the AlexNet architecture and its pre-trained weights prior to training on the client side.

Federated Learning could be deployed as a privacy preserving approach in detection of DR. It enhances the privacy of data compared to the traditional machine learning. Federated learning uses an averaging function, which averages the weights collected from each clients to build a global model without accessing the data. This whole concept is depicted in Fig. 1. The basis behind the two deployed federated learning approaches are the same. The only difference is in their aggregation strategies. FedProx achieves better results when facing heterogenous data. Basically, FedProx can be a more applicable approach of FedAVG. It uses a proximal term(μ) to regularize the calculated loss. For all the experiments in this research study, we assigned the proximal term to 0.1 in our FedProx models.

C. Datasets

This study was performed on five publicly available Diabetic Retinopathy datasets including EyePACS [14], Messidor [17], IDRID [39], APTOS [23], and University of Auckland (UoA) [40]. Most of these datasets contain five classes of diabetic retinopathy disease levels, including normal, mild, moderate, severe, and proliferative. In order to achieve more accurate and efficient DR detection, we exclude the grades of mild and moderate from our experimental datasets. Additionally, we combine severe and proliferative since detecting both is critical to refer the patients to the doctor. This will also lower the imbalance between two classes in the datasets. There are more normal retina images than critical levels. As a result, two classes of normal and critical are created for each dataset as depicted in Fig. 2.

TABLE I
NUMBER OF SAMPLES FOR DATASETS

Dataset	Classes		Total
	Normal	*Critical*	
EyePACS [14]	673	521	1194
Messidor [17]	468	110	578
IDRID [39]	168	144	312
APTOS [23]	283	420	703
UoA [40]	56	85	141

Our first dataset, EyePACS, contains many samples compared to others. This dataset was published for a Kaggle competition. A total of 1194 images from this dataset are utilized for our experiments. Messidor is the most unbalanced DR dataset among the others. The number of images adopted from Messidor is 560. However, we also added the images for critical grade from the new version of the Messidor dataset to our collection. This is performed to reduce the overfitting due to the unbalanced classes. Consequently, the total images from the Messidor dataset are 578. Another large DR dataset is the APTOS, containing 703 images of two classes. The last two datasets deployed in this study are IDRID and UoA, which have 312 and 141 images, respectively. Table I and Fig. 3 visually demonstrate the number of samples for each DR class in the datasets mentioned above.

D. Preprocessing

Since different institutions collected the images, we cropped all the images to the same size. Additionally, we utilized some augmentation techniques to avoid over-fitting and bias within our models. Since it is essential to have the whole retina image for the experiments, we only applied the random horizontal flip on the retina images. We have written and executed our code within PyTorch (version 1.4.0) and PySyft (version 0.2.9) using a GPU of Tesla M40 with a 12 GB memory size and 288.4 GB/s bandwidth for all of the experiments. Our evaluation metrics for these experiments include accuracy, sensitivity, specificity, and precision. For simplicity, we demonstrate our results by comparing the achieved prediction accuracies.

IV. RESULTS AND DISCUSSIONS

This section represents the achieved results of our three models for detecting diabetic retinopathy. We first demonstrate the individual training and testing results on each of the five datasets. Second, we deploy cross dataset testing on our best-performed dataset within the first scenario. We train the datasets using all of their samples and test them using a different dataset.

TABLE II
METRICS FOR SCENARIO 1

Methods	Datasets	Metrics			
		Acc. %	*Sen. %*	*Spec. %*	*Prec. %*
Standard	EyePACS	81.66	92.64	67.30	78.75
	Messidor	87.93	100	36.36	87.03
	IDRID	93.75	100	86.66	89.47
	APTOS	91.54	86.20	95.23	92.59
	UoA	100	100	100	100
FedAVG	EyePACS	78.33	98.52	51.92	72.82
	Messidor	81.03	100	0	81.03
	IDRID	87.50	88.23	86.66	88.23
	APTOS	85.91	86.20	85.71	80.64
	UoA	100	100	100	100
FedProx	EyePACS	82.5	91.17	69.23	79.48
	Messidor	87.93	97.87	45.45	88.46
	IDRID	90.62	94.11	80	84.21
	APTOS	91.2	86.20	85.71	80.64
	UoA	100	100	100	100

A. Scenario I

Standard, FedAVG, and FedProx models are trained and tested on each dataset. The UoA dataset achieves the best accuracy of Standard models with 100% correct predictions. Next, IDRID achieves an accuracy of 93.75%. Interestingly, FedAVG and FedProx also achieve 100% accuracy on the UoA dataset. We can see that the models correctly predicted all the classes by checking their confusion matrix.

Additionally, the best-performed datasets after UoA for FedAVG is the IDRID showing an accuracy of 87.5%, and for FedProx is the APTOS with an accuracy of 91.2%. Complete details of the models' performances for individual training and testing are demonstrated in Tablel II. We can see that FedProx outperforms the results achieved by FedAVG in this scenario, which is due to the excellent performance of FedProx on the heterogeneous datasets.

B. Scenario II

Since the UoA dataset has outperformed all the other datasets by achieving the highest accuracy, we assign the UoA as our testing dataset in this scenario. In this case, each dataset is trained with its data and then is evaluated with the UoA dataset. The Standard model attains the highest accuracy of 92.19% and 90.78% for APTOS and IDRID with different batch sizes per iteration. FedAVG also demonstrates excellent performance on the APTOS dataset with an accuracy of 90.07%. It also performs well on the IDRID with 86.52% prediction accuracy.

The last model, FedProx, attains the accuracy of 85.81% and 82.97% on APTOS and Messidor. We can conclude that maintaining high-quality images with a balanced number of samples for each class could highly increase models' output. The performance of all four datasets tested on the UoA dataset is shown in Table III. Fig. 4 depicts the accuracy of each model for this scenario.

FedAVG and FedProx results dropped compared to the first scenario due to the highly heterogeneous datasets, especially the EyePACS. Additionally, the training time for two models of

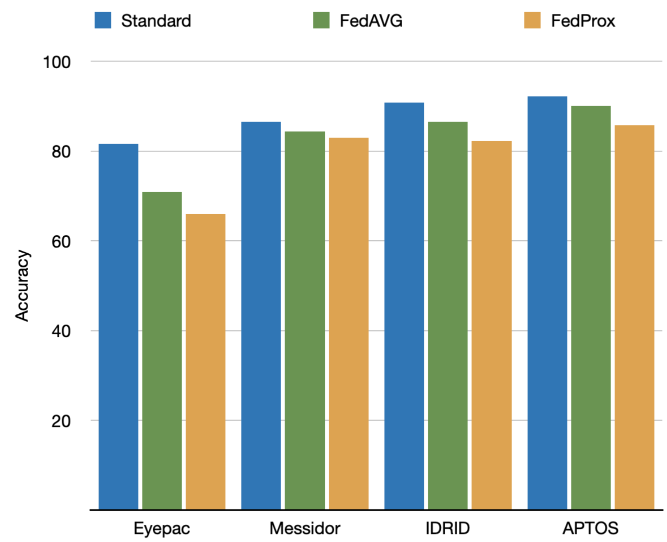

Fig. 4. Performance of different models trained on different datasets and tested on UoA

FedAVG and FedProx is higher than the standard model, which has been a limitation of applying FL in a decentralized manner compared to the traditional ML training. Based on the achieved performances, we should mention that our experiments using individual and cross datasets have outperformed the conducted works discussed earlier in the related work section.

TABLE III
METRICS FOR SCENARIO 2

Methods	Datasets	Metrics			
		Acc. %	*Sen. %*	*Spec. %*	*Prec. %*
Standard	EyePACS	81.56	53.57	100	100
	Messidor	86.52	98.21	78.82	75.34
	IDRID	90.78	87.5	92.94	89.09
	APTOS	92.19	91.07	92.94	89.47
FedAVG	EyePACS	70.92	26.78	100	100
	Messidor	84.39	100	74.11	71.79
	IDRID	86.52	94.64	81.17	76.81
	APTOS	90.07	85.71	92.94	88.88
FedProx	EyePACS	65.95	14.28	100	100
	Messidor	82.97	80.35	84.70	77.58
	IDRID	82.26	91.07	76.47	71.83
	APTOS	85.81	67.85	97.64	95

V. CONCLUSION

This paper demonstrated the efficient use of a pre-trained network called AlexNet within three different approaches. The main idea for employing AlexNet was to detect diabetic retinopathy in the retinal images. To train deep networks using transfer learning, the models should access the data. However, this study showed that FL aggregation strategies could achieve different performance and computing times without accessing the data. Our approach outperformed all the previously conducted studies in terms of prediction accuracy. In addition, deploying FL enabled us to simulate real-time training data in the medical context, where accessing data is not allowed due to the regulatory privacy rules. However, FL still faces different challenges and issues. We aim to deploy a simulation based on federated collaborative learning for our future work, where each client acts as a hospital containing different data. Moreover, we will apply differential privacy on our FL setup to increase aggregated parameter privacy.

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